Problem Definition and Solution Strategy Write-Up

Project Objective

- Current state of the world is a manual process where a human is validating all forms of identification and cross checking to confirm if a passenger is on the correct flight.
- Costs of human involvement in the process include:
  - Time
  - Efficiency
  - Errors in Validation
  - Cost of Labor
- This process is easily automated using Azure AI solutions
- Ideal state using AI solutions, the strategy will include:
  - Passengers scanning their ID and boarding pass while the azure solution extracts that information
  - Facial recognition is performed to gather passenger sentiment
  - Prohibited item scanning is done at the kiosk as well
  - The kiosk should be able to give the passenger a final message that lets them know whether or not they have passed validation and can board the plan

Data Elements

1. Flight Manifest List for all 5 passengers
2. Passenger ID Card
3. Passenger Boarding Pass
4. Passenger video showing their face for sentiment analysis
5. Passenger carry-on items photo

Solution Strategy

- The base of it all is Azure Blob Storage which will store all the data
- Then Azure Document Intelligence will extract the passenger information from the boarding pass as well as the Digital ID and cross check the data to confirm the passenger's identity
- Azure Custom Vision will assist with pulling the ID photo and the images of the contraband by training it on data of various images of that contraband
- The Video Indexer will be used for the passenger video at the kiosk in order to verify their identity with their ID
- The Face API will also use the video to extract sentiment
- Finally all of this will be used to come to a binary conclusion based on the manifest list whether or not the passenger passed verification and can be boarded